

SECTION 15130-A

PUMPS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. All pumps except where integral with a manufactured piece of equipment.
- B. Pump controls where contained within a control panel assembly.
- C. Manufacturer shall ensure that all equipment provided meets the dynamic balance and vibration isolation requirements detailed in Section 15240 Mechanical Vibration Control.

1.3 SUBMITTALS

- A. Submit shop drawings and product data in accordance with Division 1.
- B. Submittals shall include the following:
 - 1. General:
 - a. Input data for selection.
 - b. Model number of the unit.
 - c. Outline specification indicating materials and other pertinent information. Submit system start-up, testing and performance certification.
 - d. Submit data sheets.
 - 2. Shop drawings:
 - a. Dimensioned plans and elevation drawings.
 - b. Dimensioned component locations and all clearances including maintenance access areas.
 - c. All weights.
 - d. Dimensioned support points.
 - e. Dimensioned utility connection points.
 - 3. Utilities:
 - a. Indicate the quality and quantity of each utility.
 - b. Indicate size and location of all utility connection points.

- c. Single line schematic Drawing of the power and utility field hookup requirements. Indicate all items furnished.
- 4. Components and performance:
 - a. Locate and indicate all components.
 - b. Provide manufacturer's literature and performance data.
 - c. Provide sound power levels data generated from the pump casings at rated capacity in 8 octave bands.
 - d. Certified pump curves showing pump performance characteristics with pump and system operating point plotted. Include NPSH curve when applicable.
 - e. Detail Drawings showing dimensions and piping connections.
 - f. Electrical data, including motor starting requirements.
 - g. Manufacturer's installation instructions.
 - h. Operation and Maintenance manuals, including schedule for lubrication and recommended preventative maintenance.
 - i. Millwright pump alignment certification.
 - j. Manufacturer's start-up report, including performance verification.
 - k. Dynamic balance report.
 - l. Drawings, cut sheets and calculations showing vibration isolation hardware selections and installation, if required by Section 15240.
- 5. Controls:
 - a. Schematic diagram of the control system. Indicate points of field interface/connection.
 - b. Schematic diagram shall indicate field installed and factory wiring.
- 6. Handling, Installation, Operation and Maintenance Data:
 - a. Submit installation and rigging instruction.
 - b. Submit sequence of installation and utilities connection as required.
 - c. Include start-up instructions, maintenance data for the controls and all components. Include trouble shooting guide.
 - d. Submit operation O&M data to include the above information.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. All pump manufacturers must have local agents to provide service after installation.
- B. HVAC Pumps:
 - 1. According to the recommended manufacturer table.
- C. Fire Pumps:
 - 1. According to the recommended manufacturer table.
- D. PCW Pumps:
 - 1. According to the recommended manufacturer table.

2.2 MOTORS AND CONTROLS

- A. Motors: Provide with equipment in accordance with Section 15010.
- B. Fire Pump Motors: Listed specifically for fire pump applications.
- C. Motor Starters: Provided under Division 6 as referred to in Section 16440 except when provided within equipment control panels, or part of fire pump controllers.

2.3 GENERAL

- A. All rotating parts shall be statically and dynamically balanced as specified in Section 15240 Mechanical Vibration Control.
- B. Especially designed to suit temperatures involved when handling hot liquids.
- C. Size and capacity scheduled on Drawings.
- D. Balance: All rotating equipment must not exceed maximum allowable vibration as specified in Section 15240 Mechanical Vibration Control.
- E. Pumps shall be selected such that the rated condition point shall lie at or to the left of the best efficiency point on the pump performance curve but shall not be at a point representing less than 20 percent of rated capacity for the selected impeller size.
- F. Impellers shall be sized so that the NPSHR is at least 2 feet less than NPSHA at rated capacity. Pumps with

constant speed drives shall be capable of at least a 15 percent increase in head at rated conditions by replacement of the impeller.

- G. For pumps intended for parallel operation pump performance curves shall increase continuously to shutoff and one pump operating alone shall not operate beyond its performance curve.
- H. Bearing materials shall be recommended and guaranteed by the pump manufacturer. Bearing life shall be calculated in accordance with ANSI/AFBMA 9, Load ratings and Fatigue life for roller bearings, and guaranteed for a minimum life of 30,000 hours at rated conditions
- I. Impellers shall be balanced in accordance with ISO G6.3.
- J. Pumps shall have spacer-type shaft couplings unless otherwise specified.
- K. Shafts (Not applicable to vertical submerged or vertical turbine pumps):
 - 1. Only solid shafts are allowed.
 - 2. Dynamic shaft deflection at the impeller centerline under maximum load shall not exceed 0.002 inch at the face of the mechanical seal.
 - 3. Shaft runout at the face of the mechanical seal and at the impeller shall not exceed 0.002 inch.
 - 4. Runout at the face of the stuffing box (or seal chamber) shall not exceed 0.002 inch.
 - 5. Stuffing box (or seal chamber) register runout shall not exceed 0.002 inch.

2.4 BASE-MOUNTED END SUCTION PUMPS

- A. Pumps: Centrifugal, single stage, integrally mounted with motor upon a fully enclosed cast iron or reinforced steel base. A flexible type coupler with guard, shall connect the motor to the pump. Coupler to be as manufactured by Woods, Lovejoy or equivalent.
- B. Casing: Minimally rated at 175 psi working pressure, furnished with gauge, vent, and drain ports. Class 30 cast iron volute with integral pedestal support. Bronze, fully enclosed impeller, keyed to shaft and cap screw secured.
- C. Mechanical Seal: Suitable for continuous operation at 225 degrees F, provided with a ceramic seal seat, carbon seal ring and replaceable bronze shaft seal.

- D. Bearings: Re-greaseable, heavy duty ball bearings, replaceable without disturbing piping connections.
- E. Suction Diffuser: Pump and system connections will be at right angles to each other. Assembly will be suitable for working pressures to 175 psig and temperatures to 250 degrees F. Diffuser body will be cast-iron or fabricated steel with stainless steel combination diffuser-strainer-flow straightener assembly and disposable brass mesh start-up strainer. Strainer core will have 3/16 inch openings with a minimum free area of 5 times the area of the pump side opening. The length of the flow straightening vane shall be at least 2-1/2 times the diameter of the pump connection.

2.5 IN-LINE PUMPS

- A. Pumps: Centrifugal, single stage, close coupled to motor, in-line, serviceable without removing from piping.
- B. Casing: Class 30 cast iron, 175 psi working pressure, provided with flanged suction and discharge nozzles, and gauge, vent, and drain ports.
- C. Impeller: Bronze, fully enclosed, dynamically balanced, keyed to steel shaft and locknut secured. Internally flushed mechanical seal with ceramic seal seat and carbon seal ring. Cover the wetted area under the seal with a nonferrous shaft seat.
- D. Bearing Bracket Assembly: Includes an oil lubricated bronze journal and thrust bearings, alloy steel shaft, and a flexible coupling.

2.6 IN-LINE CIRCULATORS

- A. Circulators: Centrifugal, single stage, flexibly close coupled, in-line, serviceable without removing from piping.
- B. Casing: Cast iron, rated for 125 psi working pressure. Impellers shall be cast iron or cadmium plated steel with a heat treated carbon steel shaft.
- C. Mechanical Seal: Carbon against a ceramic seat.
- D. Bearings: Bronze, sleeve type, oil lubricated.

2.7 DOMESTIC HOT WATER CIRCULATORS

- A. Circulators: Centrifugal, single stage, flexibly close coupled, in-line, serviceable without removing from piping.
- B. Construction: All bronze, including casing, impeller, pump bearings, etc., especially designed for domestic water service.
- C. Mechanical Seal: Carbon against a ceramic seat.
- D. Bearings: Bronze, sleeve type, oil lubricated.

2.8 SUBMERSIBLE SUMP PUMPS

- A. Pumps: Fully submersible, furnished with cast-iron housing, integral diaphragm switch, oil filled continuous-duty double ball bearing motor with built-in overload protection, corrosion protected steel shaft, non-clog bronze impeller, motor cord, and cap.
- B. Alarms: Provide separate level switch to alarm bell indicating high water level in sump. A set of normally open and normally closed alarm contacts shall be provided for remote alarm capability. Mercury switches shall be sealed in polypropylene floats and furnished with PVC coated UL listed cable, and bracketed to a galvanized suspension rod. Switch shall energize alarm system in the event of high water.
- C. Pump Basin: Cast-in-place concrete, sized as indicated on Drawings. Basin depth shall be at least 3 feet deeper than lowest inlet.
- D. Check Valve: Provide in discharge line.

2.9 PROCESS COOLING WATER PUMPS

- A. General: Centrifugal pumps shall be statically and dynamically balanced, designed to meet the requirements of Section 15240 Mechanical Vibration Control. Wetted metallic parts shall be passivated 316L stainless steel, 150 grit interior finish. Wetted seal components shall be 316 stainless steel with ceramic seat. Seals to be hydraulically balanced type. Motors to be TEFC type.
- B. Motor Bearings:

1. Bearings shall be shielded, re-greaseable, vacuum degassed steel ball bearings, specially selected for electric motor service, and longer service life.
2. Bearings shall be lubricated with a premium moisture resistant grease formulated to operate over a temperature range of minus 20 degrees F and plus 300 degrees F.
3. Bearing identification by AFBMA number shall be shown on the motor nameplate.
4. Corrosion-resistant cadmium-plated grease plugs shall be provided for relubrication of the bearings.
5. Motors including their mountings, face plates, and shaft sizes to be built to NEMA Standard dimensions and with UL approved terminal boxes. Motors shall have plugged tapping at the top and bottom of the bearing housing for proper changing of bearing lubrication.
6. Motors shall be provided with Variable Frequency Drives that include built-in DC banks for continued operation in the event of a power outage. DC banks shall be sized for ≥ 90 sec, operation at full load.

2.10 VERTICAL TURBINE PUMPS

A. Type:

1. Electric motor driven, vertical, centrifugal, turbine, factory assembled and tested.

B. Casing:

1. Cast iron, ANSI rated for 125 psig, provided with discharge gauge, air vent, wear rings, seal flush connection, drain plug, and flanged discharge. Top of head shall have a machined register to fit the motor. A separate cast iron or steel sole plate shall be supplied.
2. Baseplate: Furnish pump with a steel baseplate of adequate size and thickness to support the pump head, column, bowls and motor.

C. Impellers:

1. Bronze, fully enclosed, keyed to shaft or secured with locknut.

D. Shaft:

1. Stainless steel with bronze or stainless steel sleeve through seal chamber.
2. Shaft shall be sufficient diameter to transmit the pump horsepower and rigidly support the impellers between the bowl or case bearings. Bowl shaft

material shall be high chrome stainless steel ASTM A 276, Grade 410.

- E. Seals:
 - 1. Mechanical seal.
- F. Column Pipe:
 - 1. Sized per AWWA-E101, with flanged connections both ends. Column length as indicated on the pump schedule.
- G. Bowl Assembly:
 - 1. Cast iron pump bowl and suction bell assembly. Minimum two bearings above the impeller shall be provided. Bearings shall be bronze.
 - 2. Bushing: Bowls and cases shall have bronze sleeve type bushings to support the shaft. Bronze alloy shall be ASTM B 505-836.
 - 3. Suction Case Bearing: Suction bearing shall be grease packed for permanent lubrication. The bearing housing must be designed to allow bearing removal by use of a press without damage. A sand collar of rubber shall be provided to protect the suction case bearing from abrasives in the liquid pump.
- H. Non-Reverse Ratchet. Shall be provided to prevent the pump from turning the motor.
- I. Motor:
 - 1. NEMA rated induction motor, TEFC, with vertical shaft, high thrust type rated for continuous duty, suitable for 460V/3-phase/60 Hz, anti-friction grease lubricated thrust bearing, and shall have an AFBMA minimum life rating of 5 years based on the maximum thrust at any specified operating condition.
 - 2. Motor shall be constructed so that the total hydraulic and static thrust of the pump's assembly can be carried by the motor thrust bearing(s). Motor shall be mounted directly on the pump discharge head assembly within a registered fit for correct shaft alignment.
 - 3. Nameplate rating of the motor shall not be exceeded at any operating condition.

2.11 FIRE PUMP SYSTEM

- A. General: Fire pump, jockey pump, motors, engines, controller, and related accessories shall be provided fully packaged, factory piped, wired and tested, on structural steel base, in compliance with FM Data Sheet

3-7N, NFPA 20, and shall be UL listed and FM approved. Refer to schedule on Drawings for sizes and capacities.

1. Components: Provide components listed in the FM Approval Guide.
2. Electrical:
 - a. System Power: Provide single point 460V/3PH/60HZ connection with fused disconnect switch.
 - b. Control Power: Provide system control power transformer.
 - c. Fire Alarm System Connections: Remote alarms at panel terminal strips and valve taper switches at valves. Coordinate characteristics with Division 16.
 - d. Wiring and Raceway: Comply with NFPA 20 and Division 16 requirements.
3. Inlet and Exhaust System:
 - a. Inlet air Filter: Dry Type
 - b. Spark Arrestor/Silencer: Residential noise rated, heavy industrial type with flexible exhaust connection, ANSI flanges.
 - c. Sound power level: 45 dBA at 35 meters.
 - d. Exhaust manifold: water cooled.
4. Seismic Protection Criteria: Design equipment, piping, wiring, and accessories anchorage and bracing systems to comply with specification 15245 Mechanical Seismic Control.. Provide fire protection system piping sway bracing per NFPA 13 based on a horizontal force factor of 50% of the weight of water filled piping.
5. Additional Submittal Requirements:
 - a. Anchorage and Bracing Shop Drawings: Submit detailed drawings showing proposed anchors and bracing to be provided. Drawings shall be stamped and signed by a Professional and shall include project name and drawing number.
 - b. Anchorage and Bracing Calculations: Submit seismic anchorage and bracing structural calculations signed by Vendor's licensed engineer. Vendor's licensed engineer shall be responsible for compliance with specified design criteria.
 - c. A certificate on calculation cover sheet that states: "These calculations demonstrate that system detailed complies with requirements of applicable local building codes.
 - d. Calculations, stamped and signed by a Professional.
6. Quality Assurance:
 - a. Engineer Qualifications: Registered professional engineer licensed to practice structural engineering in jurisdiction where Project is located.

B. Design:

1. Horizontal, split case, cast iron, bronze fitted, centrifugal fire pump approved by Factory Mutual. Unit shall comply with FM Data Sheet 3-7N, and National Fire Protection Association Pamphlet No. 20. Pumps shall be mounted on a fabricated steel base, complete with coupling, coupling guard, and directly connected to motor.
2. Pump discharge flange shall be rated for 250 lbs.
3. Impeller: Bronze, fully enclosed, statically and dynamically balanced, keyed to shaft. Provide annular type cast iron casing wearing rings, locked against rotation and cast bronze impeller wear ring integral with the impeller.
4. Steel Shaft: Fully machined, keyed at coupling end, provided with renewable bronze or stainless steel shaft sleeves which shall be firmly butted against the impeller hub with threaded sleeve nuts. Ring type packing shall be provided along with a split water seal cage. Provide lubricated deep grooved ball bearings on the shaft inboard and outboard radial areas.
5. Fittings and Accessories:
 - a. 1/2 inch automatic air release valve.
 - b. Suction and discharge pressure gauges.
 - c. 3/4 inch casing relief valve.
 - d. 10 inch by 8 inch eccentric suction reducer (250 lb. cast iron) for booster pump.
 - e. 12 inch by 8 inch eccentric suction reducer (250 lb. cast iron) for suction tank pump.
 - f. 6 inch by 8 inch concentric discharge increaser (250 lb. cast iron).
 - g. 8 inch by 8 inch discharge tee (250 lb. cast iron).
 - h. Related valves and piping.

C. Casing: Cast iron, dowelled, single volute, axially split, with flat face suction and discharge flanges cast into the lower half, and integral pedestal support. Flanges rated at 150 psi or 1.25 times actual working pressure, whichever is greater. The 2 halves shall be bolted together and designed for removal of shaft assembly without disturbing piping or motor mounting. Provide tappings for sealing fluid at stuffing box area, drain holes, air release or volute priming, suction and discharge gauges. Provide split glands with swing bolts as a packing retainer in the stuffing boxes.

D. Drivers:

1. Motors: FM approved for fire pump use; 1750 RPM, 3-phase, 60 cycle, 460 volt, horizontal open drip-proof motor with a 1.15 service factor.
2. Motors shall be suitable for auto transformer soft start operation.
3. Engines: Diesel engine shall be UL listed and FM approved and shall be arranged for automatic start and manual stop. Diesel engine cooling shall be by a heat exchanger receiving water from the discharge of the pump, and completed with an automatic electric solenoid valve, pressure regulator strainers and by-pass piping. The engine shall be equipped with, but not limited to the following:
 - a. Fuel injection system consisting of the fuel injectors, fuel pump, strainer, filter and the necessary connecting fuel lines.
 - b. Air inlet cleaner
 - c. Lubrication oil system consisting of an oil pump, oil cooler and filter with by-pass valves.
 - d. Jacket water heater.
 - e. An instrument panel consisting of an oil pressure gauge, ammeter, tachometer, totalizing time meter.
 - f. A governor capable of regulating engine speed within a range of 10 percent between shutoff and maximum load conditions of the pump. The governor shall be field adjustable and set to maintain the rated pump speed at maximum load.
 - g. Manually operated shutdown device.
 - h. Electric starting motor with overrunning clutch drive and starter switch.
 - i. Muffler with metallic flexible connector to the engine.
 - j. Overspeed switch, low oil pressure switch, dual starter contactors, all wired to junction box for direct connection to a UL/FM approved dual battery automatic engine controller.
 - k. Flexible shaft coupling
 - l. Flexible metallic connections between cooling water lines and the engine.
 - m. One set of dual batteries, lead-acid storage heavy duty type with battery rack and cables for starting the diesel engine driven pump, the voltage shall meet the engine requirements. 110 volt single phase built in battery charger shall be suitable for the dual battery system. Each battery unit shall have capacity at 4.5 degrees C sufficient to maintain cranking speed recommended by the engine manufacturer through a six minute cycle. The charger shall be suitable for keeping the engine starting batteries charged during the

periods when the engine is idle. The charger shall be equipped with protective relays to be interlocked with the engine starting circuit to disconnect the alternating current supply to the charger during the engine cranking cycle.

E. Fire Pump Controllers:

1. Fire pump controllers shall be a single factory wired unit meeting the requirements of the most recently mandated edition of FM Data Sheet 3-7N, and NFPA 20, UL listed and labeled , approved by Factory Mutual and labeled for fire pump service, NEMA 2 cabinet containing a single handle for operating both an isolating switch and circuit breaker in sequence, starting contactor, built-in pressure switch for automatic starting, manual start and stop pushbuttons, remote start circuit, dry contacts, for remote alarms of loss of primary or control power, pump running, power available pilot light, and an emergency start lever to close contactor independent of automatic control circuits.
2. Controllers shall be FM approved combined manual and automatic type designed for automatic start and manual stop, with:
 - a. The minimum withstand rating of the controller shall not be less than:
 - 1) 100,000 Amps, RMS symmetrical at 480 volts.
 - b. Controller manufacturer shall provide a 5 year warranty to include all parts and replacement labor covering the controller power circuit components, including isolating switch, circuit breaker, current sensing module, motor contactor, and PCL assembly.
 - c. Isolating disconnect switch and circuit breaker shall be mechanically interlocked so that the enclosure door cannot be opened with the handle in the "On" position except by a hidden tool operated defeater mechanism.
 - d. Controller shall include electronic annunciator with externally mounted, individual visible indicators for "Power Available", "Low Pressure", "Local Start", "Remote Start", "Phase Failure", "Phase Reversal", "Interlock On", "Pump Operating," "Failure to Start", and "Run Timer On."
 - e. Controller shall be supplied with duplicate, individual "Phase Failure", "Phase Reversal", "Motor Overload", "Failure to Start", "Low Pressure", and "Pump Operating" alarm contacts wired to Fire Alarm System (FAS) as indicated.

- f. Alarms for the diesel driver shall be provided for Overheated engine, High pump room temperature, and Low fuel.
- g. A panel mounted pressure switch shall be provided to open or close all circuits necessary to automatically start the engine by performing the following functions: Turn on starting equipment, crank the engine in a series of crank and rest cycles automatically alternating between the dual batteries on each cycle, disconnect the cranking motor upon engine start. Once the engine has started it shall remain in operation until it is manually shut down.

F. Piping and Valves:

- 1. Materials and Components: Comply with Section 15010 Mechanical General Provisions. Fuel oil piping shall comply with 15060 Pipe and Pipe Fittings - General and 15063 Steel Pipe, Fittings and Valves.
- 2. Arrangement: As indicated on Drawings.
- 3. Booster Pump:
 - a. 10" by-pass with non-slam check valve and two butterfly valves.
 - b. 10" pump suction OS&Y gate valve.
 - c. 8" pump discharge non-slam check valve.
 - d. 8" test header isolation valve.
 - e. 8" pump discharge isolation valve.
 - f. 1-1/2" jockey pump suction OS&Y gate valve.
 - g. 1-1/2" jockey pump discharge non-slam check and OS&Y gate valve.
- 4. Suction Tank Pump:
 - a. 8" pump discharge non-slam check valve.
 - b. 8" test header isolation valve.
 - c. 8" pump discharge isolation valve.
 - d. 1-1/2" jockey pump tank suction swing check and OS&Y gate valve.
 - e. 1-1/2" jockey pump domestic suction OS&Y gate valve.
 - f. 1-1/2" jockey pump discharge non-slam check and OS&Y gate valve.

G. Jockey Pumps:

- 1. Pumps: Horizontal shaft, centrifugal, close coupled to motor, cast iron casing, constructed with stainless steel impellers, chambers, and shaft seal, tungsten carbide seal surfaces, ceramic bearings., and cast iron mounting base.
- 2. Jockey Pump Controllers:

- a. Controller shall be UL listed, NEMA 2 cabinet, drip proof enclosure, fabricated from heavy gauge cold rolled steel, containing a system fused disconnect switch with external operating handle, across-the line full voltage starter with overload contacts, pressure switch with high-low set points for automatic starting, hand-off-auto switch for manual starting, minimum run period timer and control transformer.
 - b. Accessories:
 - 1) Casing relief valve.
 - 2) Running period timer.
 - 3) Suction and discharge pressure gauges.
 - 4) 115V control transformer.
 - 5) Pump on light.
 - 6) Fuses
- H. Field Acceptance And Performance Test:
- 1. Pump manufacturer's representative shall be present for the field acceptance test. Testing shall meet requirements of FM Data Sheet 3-7N, and NFPA Pamphlet 20.
- I. Factory Prefabrication:
- 1. Fire pump system shall be mounted on an open structural steel baseplate. Piping, pressure sensing lines, pump by-pass, (if required), shut off valves, stuffing box, air vent, and casing relief valve drains will be firmly anchored to the steel base by means of structural steel supports. Electrical connections and wiring will be completed and tested at the pump manufacturer's factory prior to shipment to the Job Site.
- J. Factory Test:
- 1. System shall be factory tested in accordance with the requirements of NFPA, UL, and FM. Additionally, the entire package system will be hydrostatically tested by the system manufacturer prior to shipment.
- K. Guarantee:
- 1. Fire pumping system shall be guaranteed in writing by the manufacturer for a period of 1 year from date of satisfactory field test against defect in design, material, or construction.
- L. Start-Up Service:
- 1. Services of a factory trained representative shall be made available on the Job Site for 2 days to check installation, field acceptance testing, and instruct operating personnel.

2.12 HORIZONTAL SPLIT CASE PUMPS

- A. Pumps: Centrifugal, single stage, integrally mounted with motor upon a fully enclosed cast iron or reinforced steel base. A flexible type coupler with guard, shall connect the motor to the pump. Coupler to be as manufactured by Woods, Lovejoy or equivalent.
- B. Pump Construction: Casing shall be (close-grained cast iron for working pressures up to 175 psig) and shall be of axially-split design with suction and discharge flanges and mounting feet cast integral with the lower half casing. Tapped and plugged holes shall be provided for priming, vent, drain and gauge connections. Upper half casing shall be removable without disturbing suction or discharge piping.
- C. Casing Connection: Flanges shall be 125 lb, ASA Standard. Suction and discharge shall be on a common centerline in both the horizontal and vertical planes.
- D. Impeller shall be of the enclosed double-suction type made of bronze non-overloading in operating characteristics and statically and hydraulically balanced. The impeller shall be keyed to the shaft and fixed in an axial position.
- E. Shaft shall be made of 416 stainless steel.
- F. Stuffing box housings and bearing brackets shall be made of cast iron separate from the casing and mounted in cylindrical fits in the ends of the casing.
- G. Mechanical Seal: Suitable for continuous operation at 225 degrees F, provided with a ceramic seal seat, carbon seal ring, and replaceable bronze shaft seal.
- H. Bearings: Regreaseable, heavy-duty ball bearings, replaceable without disturbing piping connections.
- I. Casing rings shall be made of bronze and shall be installed with an anti-rotation device, and designed to prevent leakage across the ring fit.
- J. Impeller rings made of bronze shall be mounted on the impeller hubs to provide for renewable clearances.
- K. Baseplate shall be channel steel (with) (without) drip rim, sufficiently rigid to support the pump and driver. Final alignment of pump and driver shall be made after

grouting and installation, and shall be approved by customer prior to operation.

- L. Coupling shall be (flexible type) with steel flanges connected by a rubber sleeve for torque transmission or (all metal type).
- M. Coupling Guard: Coupling guard shall be the all metal type.
- N. Double suction diffusers manufactured by Flow Conditioning Corporation for symmetrical flow and separation of stream debris shall be installed at the inlet of double suction pumps as indicated on plans. Diffuser shall consist of elbow type ductile iron body with exit vane, integral flanged pump connection, flanged system connection, $\frac{1}{4}$ inch gage tape, 1-1/2 inch blowdown connection and pipe support bosses located to assure no movement between pump and support base. An "O" ring sealed removable cover shall permit inspection and removal of double inlet bronze and stainless steel orifice cylinder/venturi assembly in a direction perpendicular to the axis of the pump inlet. Diffusers for closed system operation shall be equipped with a readily replaceable start-up strainer which can be removed after initial operation and preserved for future use. Orifice cylinder shall be selected to withstand a pressure differential equal to pump shutoff head.

2.13 GENERATOR FUEL TRANSFER PUMPS

- A. Pumps: Positive displacement, rotary, single stage, direct driven through flexible coupling with guard, with motor, upon reinforced steel base.
- B. Casing: Minimally rated at 175 psi working pressure, furnished with opposite ports, integral safety relief valve on head, Class 30 cast iron.
- C. Mechanical Seal: Suitable for continuous operation, rotary type package unit, self-adjusting.
- D. Bearing Bracket Assembly: Sealed, radial type ball bearing supported shaft.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide drains for bases and stuffing boxes piped to and discharging into floor drains.
- B. Provide air cock and drain connection on horizontal pump casings.
- C. Provide flexible pipe connections in accordance with Section 15240 Mechanical Vibration Control.
- D. Decrease from line size with long radius reducing elbows or reducers. Support piping adjacent to pump so no weight is carried on pump casings. Provide supports under elbows on pump suction and discharge line sizes 4 inches and larger.
- E. Ensure pumps operate at system fluid temperatures without vapor binding and cavitation, are non overloading in parallel or individual operation, operate within 25 percent of midpoint of published maximum efficiency curve.
- F. Provide services of qualified millwright to check, align, and certify base-mounted pumps prior to start-up.
- G. Coordinate electrical hookup.
- H. Grout bases as required by manufacturer.
- I. Mount rotating equipment on vibration isolators and vibration isolate piping and conduit as specified in Section 15240 Mechanical Vibration Control.

3.2 START-UP

- A. Activate water and chemical treatment systems.
- B. Perform waterside test and balance.
- C. Provide services of factory-trained representative to supervise testing, starting of equipment, and instructions on operation and maintenance.

- END -